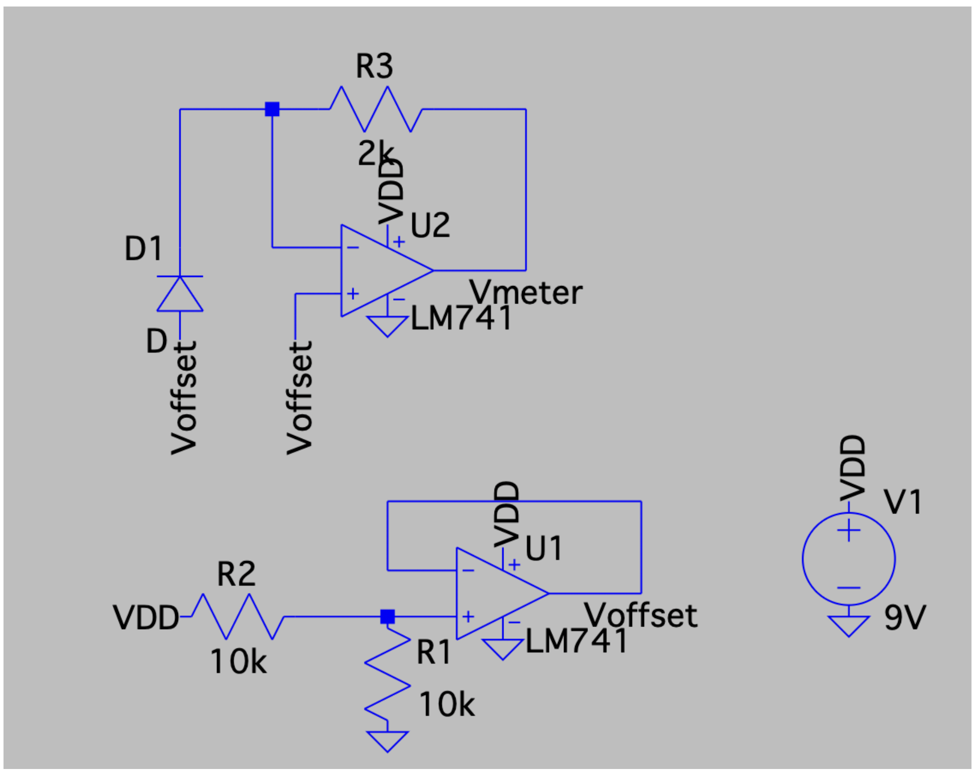
Shiyao Marcus Lam (sml2286)
Shawn Hans Kathuria (shk2199)
Electronic Circuits Lab 3

Introduction:
In this lab, we design a basic light meter using resistors, LM741 op-amps, a solar cell (photodiode), and a 9V single supply. This light meter needs to output 0-1V, where 1V indicates maximum light intensity on the solar cell.

Design:
Here is the LTSpice of our circuit design.



There are 2 core components to our circuit. The first is the opamp network on the bottom of the LTSpice image. This is a simple buffer circuit that uses a voltage divider to consistently supply Voffset = 4.5V, suggested by LA Jayden.

We then utilize this voltage offset to the second opamp network on the top of the LTSpice image.

1. Since our solar cell is supposed to output 0 to 0.5mA of current depending on the light intensity we expose it to, according to our lab spec, we use a 2Kohm feedback resistor so that the voltage drop will vary from 0 to 1V (0.5mA * 2Kohm).
2. If we simply ground the non-inverting input and the photodiode's anode, the Vmeter output voltage will theoretically output 0 to 1V. However, due to our limitation of the single 9V supply, negative opamp supply is grounded, causing the bottom-bound output

of 0V tol saturate the opamp. This is where the buffer circuit comes in: we impose the 4.5V Voffset on top of the voltage drop across the feedback resistor so that we get a stable Vmeter output.

3. For our final reading, we probe Vmeter with Voffset as the reference voltage. This rids the 4.5V offset from our light meter output, giving us our 0-1V range.

Results:
After building the circuit, we measured Vmeter with Voffset as the reference voltage when we cover vs. shine a phone flashlight onto the solar cell.

When the solar cell is covered, our light meter outputs ~0V. When we shine the phone flashlight onto the solar cell, our light meter outputs 0.62V, so not the 1V upper-bound that we expected. This may be attributed to the following possibilities:

- We may not have exposed the solar cell to enough light intensity. The phone flashlight might not be a strong enough light source to drive the cell to its maximum rated current of 0.5mA, resulting in the lower output of 0.62V.
- The solar cell could be faulty, meaning its actual maximum current output capacity is less than the 0.5mA specified in the lab, even under maximum light conditions.